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Paper 1: Pharmacokinetic Drug–Drug Interaction Potential of Oral Anti-cancer Drugs

Bibliographic Information

- **Title:** Pharmacokinetic Drug–Drug Interaction Potential of Oral Anticancer Drugs
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Paper Type: GENERIC

Quick Take

This study systematically compiled pharmacokinetic drug–drug interaction (DDI) mechanisms for 99 oral anticancer drugs (OADs) from FDA labels and primary literature, finding that ~86% have at least one clinically relevant DDI mechanism. In a retrospective cohort of 3,697 advanced solid cancer patients, 17.4% had a potential DDI, most commonly involving CYP3A perpetrators and CYP2D6 perpetrators.

Executive Summary

Alhurayri et al. present a comprehensive analysis of pharmacokinetic DDI mechanisms for 99 FDA-approved oral anticancer drugs (OADs) through January 2023, integrating data from drug labels and 239 primary literature studies. They identified 748 drug–drug pairs and found that 85.9% of OADs had at least one DDI mechanism, with CYP3A-related interactions being most prevalent. Using these mechanisms, they assessed potential DDI (PDDI) prevalence in 3,697 advanced solid cancer patients from a molecular tumor board cohort; 17.4% of patients prescribed an OAD had at least one PDDI. The study also identified 16 DDI mechanisms lacking actionable FDA recommendations and provided therapeutic substitution tables to mitigate DDI risk within drug classes. This work serves as a practical reference for clinicians to identify and manage clinically relevant DDIs in oncology.

Scientific Context & Motivation

Drug–drug interactions (DDIs) are a major concern in oncology due to the narrow therapeutic indices of anticancer drugs and the high prevalence of polypharmacy in cancer patients. Oral anticancer drugs (OADs), particularly targeted therapies, are increasingly used but carry significant DDI risk via CYP enzymes and drug transporters. Prior retrospective studies reported highly variable PDDI prevalence (5–46%) depending on the database used, highlighting the need for standardized, evidence-based DDI classification. This study addresses the gap by systematically curating clinically relevant PK DDI mechanisms from FDA labels and primary literature, and applying them to a real-world cohort to estimate PDDI prevalence.

✂ Methodological Snapshot

The study had two components. First, DDI mechanisms were identified for 99 OADs approved through January 2023 by reviewing FDA drug labels (original and most recent) and conducting a PubMed search for clinical DDI studies, PBPK analyses, POPPK analyses, and health outcomes studies. Inclusion required ≥ 2 -fold AUC change (moderate/strong) or significant health outcomes. Data from 239 studies yielded 748 drug–drug pairs. DDI mechanisms were assigned for CYP enzymes, transporters, acid-reducing agents, and specific drugs. FDA label recommendations were categorized as contraindicated, not recommended, dose adjustment, use with caution, or no adjustment. Second, PDDI prevalence was assessed in a retrospective cohort of 3,697 advanced solid cancer patients from the Indiana University Health Precision Genomics Program (2013–2020). Patients with ≥ 1 OAD prescription were identified, and overlapping prescriptions of OADs with interacting drugs (based on the compiled mechanisms and FDA/FlockHart tables) were considered PDDIs. Days’ supply was estimated using standardized assumptions. Statistical comparisons used t-tests, Wilcoxon rank-sum, chi-square, and logistic regression.

📋 Detailed Analysis

This study’s major strength is its systematic, evidence-based approach to DDI classification, which addresses the well-documented variability among commercial DDI databases. By requiring ≥ 2 -fold AUC change or significant health outcomes, the authors likely captured only clinically meaningful interactions, explaining the lower PDDI prevalence (17.4%) compared to prior studies (26–46%). The inclusion of PBPK and POPPK studies is forward-thinking, as these methods are increasingly used in drug development and regulatory decisions. The identification of 16 DDI mechanisms without actionable FDA recommendations is concerning; for example, moderate CYP3A inducers like efavirenz can reduce gefitinib AUC by 76%, yet labels only warn against strong inducers. This suggests that clinicians should consider moderate inducers as well, especially in patients on multiple medications. The therapeutic substitution tables are a highlight, offering practical alternatives within classes (e.g., using abemaciclib instead of palbociclib/ribociclib in patients on acid reducers). However, these suggestions are theoretical and do not account for differences in efficacy, toxicity, or patient-specific factors like organ function or prior therapy. The cohort analysis revealed that prostate cancer and soft tissue sarcoma patients had the highest PDDI rates, likely due to the DDI-prone drugs used (enzalutamide, pazopanib, abiraterone) and high comorbidity burden. The multivariable analysis showing only tumor type as independently associated with PDDIs suggests that drug selection, not demographics, drives risk. Limitations include the single-center design, assumptions about days’ supply, and lack of clinical outcome data. Despite these, the study provides a valuable resource for DDI management in oncology.

Domain Context

This work sits at the intersection of clinical pharmacology, oncology, and pharmacoepidemiology. It addresses the practical challenge of managing DDIs in an era of increasing OAD use, where targeted therapies have largely replaced conventional chemotherapy. The focus on pharmacokinetic mechanisms is appropriate given their direct impact on drug exposure and the availability

of mechanistic data from drug development. The study also highlights the evolving role of PBPK modeling in DDI assessment, which is increasingly accepted by regulators. The therapeutic substitution concept leverages the fact that many OAD classes have multiple agents with overlapping indications but different DDI profiles, allowing personalized selection. This aligns with the broader trend toward precision medicine, where treatment decisions consider not only tumor genomics but also patient-specific factors like concomitant medications.

☒ Key Findings

1. Of 99 OADs, 85 (85.9%) had ≥ 1 DDI mechanism. The most common mechanisms were victims with metabolic inducers (71.7%), CYP3A substrates (55.6%), CYP3A perpetrators (29.3%), and victims with acid-reducing agents (17.2%).
 2. The primary literature search identified 16 DDI mechanisms (for 14 drugs) with ≥ 2 -fold AUC change but no actionable FDA recommendation, including moderate CYP3A inducer interactions with gefitinib, ribociclib, and ibrutinib.
 3. Within therapeutic classes, DDI mechanisms varied considerably; for 83.5% of clinically relevant DDI mechanisms, an alternative OAD in the same class without that mechanism was available.
 4. In the cohort of 1,733 patients prescribed ≥ 1 OAD, 302 (17.4%) had at least one PDDI. The most common OAD DDI mechanisms in these patients were CYP3A perpetrators (58.9%), CYP2D6 perpetrators (32.5%), and CYP3A substrates (18.9%).
 5. Patients with prostate cancer (28.1% of PDDI group) and soft tissue sarcoma (15.6%) had disproportionately high PDDI rates (32.8% and 44.8%, respectively).
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Strengths & Limitations

Strengths

- Comprehensive curation of DDI mechanisms for all 99 FDA-approved OADs using both drug labels and primary literature (239 studies, 748 drug pairs).
- Use of rigorous, clinically relevant criteria (≥ 2 -fold AUC change or significant health outcomes) to define DDI mechanisms, excluding less reliable sources like case reports.
- Inclusion of PBPK and POPPK modeling studies alongside clinical DDI studies.
- Practical therapeutic substitution tables organized by OAD class and indication, providing actionable guidance for clinicians.
- Real-world PDDI prevalence assessment in a large cohort ($n=3,697$) with detailed patient-level data.

Limitations (Acknowledged by Authors)

- Cohort limited to advanced solid cancer patients, excluding hematologic malignancies and limiting assessment of OADs with both indications.
- PDDI prevalence based on prescription orders from EHR, not capturing medication adherence or over-the-counter medications.
- Data only available until October 2020; OADs approved after that date not included in prevalence analysis.
- Only PDDIs (temporal overlap) assessed, not actual clinical outcomes.

- Assumptions about days' supply for prescriptions may introduce misclassification.

Limitations (Expert Review)

- Single-center study from a specialized molecular tumor board; results may not generalize to community oncology settings.
- Focus exclusively on pharmacokinetic DDIs; pharmacodynamic interactions (e.g., QT prolongation, bleeding risk) not addressed.
- Potential for missing interactions with weak inhibitors/inducers or those involving less common enzymes/transporters.
- Therapeutic substitution suggestions are theoretical and require clinical judgment; not validated in prospective studies.

Generalizability

The DDI mechanism list is broadly applicable to any patient population receiving these OADs. The PDDI prevalence estimate (17.4%) is specific to advanced solid cancer patients at a tertiary academic center with multidisciplinary DDI management; rates may differ in other settings. The findings are not directly generalizable to hematologic malignancies or early-stage cancers.

Figures & Tables

- **Figure 1:** Heat map summarizing DDI mechanisms and AUC changes for each OAD, organized by therapeutic class.
 - *Significance:* Provides a visual overview of the DDI landscape across all OADs, highlighting variability within classes and enabling rapid identification of substitution opportunities.
- **Figure 2:** Patient-level PDDI prevalence for the most commonly prescribed OADs, showing the percentage of patients with at least one PDDI for each drug.
 - *Significance:* Identifies high-risk OADs (e.g., pazopanib 48.2%, dabrafenib 41.7%) and helps prioritize DDI monitoring efforts.
- **Table 1:** Comprehensive list of DDI mechanisms for all 99 OADs, including CYP enzymes, transporters, victims with inducers, and acid-reducing agents.
 - *Significance:* Core reference table for clinicians to quickly check DDI mechanisms of any OAD.
- **Table 2:** DDI mechanisms with ≥ 2 -fold AUC change but no actionable FDA recommendation (16 mechanisms for 14 drugs).
 - *Significance:* Highlights gaps in current labeling that may require clinical attention, especially moderate inducer interactions.
- **Table 3:** Therapeutic substitution recommendations by OAD class and indication to avoid specific DDI mechanisms.
 - *Significance:* Actionable guidance for clinicians to select alternative OADs within the same class to mitigate DDI risk.
- **Table 4:** Characteristics of patients with and without PDDIs, including demographics, cancer type, comorbidities, and co-prescribed medications.
 - *Significance:* Identifies patient subgroups at higher risk for PDDIs (prostate cancer, soft tissue sarcoma, multiple comorbidities).

Code & Reproducibility Assessment

Not applicable; no custom code was provided. Statistical analyses were performed using R Studio (v4.3.1) and SAS 9.4.

Supplementary Materials

Supplementary materials include Appendix S1 (complete list of 748 drug–drug pairs), Appendix S2 (patient-level PDDI prevalence for all OADs), Table S1 (mechanism of action of OADs), Table S2 (drug transporters assessed), Table S3 (assumed days’ supply for pharmacy-dispensed medications), and Figure S1 (most common interacting drugs and their mechanisms).

Future Directions

Prospective studies evaluating clinical outcomes (e.g., toxicity, efficacy) of identified PDDIs are needed. Expanding the analysis to include newer OADs approved after January 2023 and to hematologic malignancies would enhance comprehensiveness. Development and validation of clinical decision support tools incorporating these DDI mechanisms could improve real-time DDI management.

Expert Commentary

This is a landmark compilation that fills a critical gap in DDI knowledge for oral anticancer drugs. The systematic approach using rigorous criteria (≥ 2 -fold AUC change) sets a new standard for DDI classification, moving beyond database-dependent variability. The therapeutic substitution tables are particularly valuable for busy clinicians, though they must be applied with caution given differences in efficacy, toxicity, and patient-specific factors. The finding that 16 DDI mechanisms lack actionable FDA recommendations highlights opportunities for label updates. Future work should link these PDDIs to clinical outcomes to confirm their relevance.

Bottom Line

This study provides a comprehensive, evidence-based reference for pharmacokinetic DDI mechanisms of oral anticancer drugs. Clinicians should be aware that ~86% of OADs have at least one DDI mechanism, and ~1 in 6 advanced solid cancer patients may experience a potential DDI. The therapeutic substitution tables offer practical strategies to mitigate DDI risk within drug classes, but individual clinical judgment remains essential.

☒ Figures

Inhibitors	3.97	1.26							
P-gp substrates	1.5	1.71							
BCRP substrates	1.35	1.4							
PARP Inhibitors	Niraparib	Olaparib	Rucaparib	Talazoparib					
CYP1A2 Substrates			2.55						
Strong Inducers		0.13		1.02					
Strong CYP3A Inhibitors		2.7							
CYP3A substrates		1.61	1.38						
CYP2C9 substrates			1.49						
CYP2C19 substrates			1.54						
P-gp inhibitors				1.58					
PI3K Inhibitors	Alpelisib	Duvelisib	Idelalisib	Umbralisib					
CYP3A Substrates		4.3	5.4	1.56					
Strong Inducers	0.26	0.18	0.25						
Strong CYP3A Inhibitors	1.37	4	1.79						
BCRP inhibitors	N/A								
SERM	Tamoxifen	Toremifene							
Strong Inducers	0.14	N/A							
Strong CYP3A Inhibitors		2.9							
Thalidomide Analogs	Lenalidomide	Pomalidomide	Thalidomide						
CYP 1A2 inhibitors		2.25							
TKI (ALK)	Alectinib	Brigatinib	Ceritinib	Crizotinib	Entrectinib	Lorlatinib			
Acid-Reducing Agents (PPIs)	1.22		0.24	1	0.75	1.01			
CYP3A Substrates	0.97	0.74	5.51	3.65	1.5	0.36			
P-gp Substrates					1.18	0.33			
Strong Inducers	0.27	0.2	0.3	0.18	0.21	0.15			
Strong CYP3A Inhibitors	1.75	2.01	2.9	3.16	6.04	1.49			
CYP2C9 substrates			1.50						
TKIs (BCR-ABL)	Asciminib	Bosutinib	Dasatinib	Imatinib	Nilotinib	Ponatinib			
Acid-Reducing Agents (PPIs)	0.99	0.74	0.57		0.66	0.94			
Acid-Reducing Agents (H2RA)			0.39		0.91				
Acid-Reducing Agents (antacids)			0.45	0.99	1.05				
CYP2C9 Substrates	4.14				1				
CYP3A Substrates	1.85			3.5	2.6				
Strong Inducers	0.85	0.06	0.18	0.26	0.2	0.38			
Strong CYP3A Inhibitors	1.36	8.6	4.84	1.4	3.01	1.9			
BCRP substrates		N/A		1					
TKIs (BRAF)	Dabrafenib	Encorafenib	Vemurafenib						
CYP1A2 substrates			4.7						
CYP2C8 inhibitors	1.49								
CYP2C9 substrates	0.63		1.18						
CYP3A Substrates	0.35	0.18	0.61						
Strong CYP3A Inhibitors	1.71	3	1.4						
Strong inducers	0.59	N/A	0.6						
BCRP and/or OATP1B1/3 substrate	1.7	1.6	1.29						
P-gp substrates			1.8						
TKIs (BTK)	Acalabrutinib	Ibrutinib	Zanubrutinib						
Acid-Reducing Agents (PPIs)	0.57	0.93	0.63						
Acid-Reducing Agents (H2RA)	N/A								
Acid-Reducing Agents (antacids)	0.47								
Strong Inducers	0.23	0.08	0.07						
Strong CYP3A Inhibitors	6.54	53.88	4.17						
TKIs (EGFR)	Afatinib	Dacomitinib	Erlotinib	Gefitinib	Lapatinib	Mobocertinib	Neratinib	Osimertinib	Vandetanib
Acid-Reducing Agents (PPIs)		0.71	0.53	0.65	1		0.35	1.07	0.93
Acid-Reducing Agents (H2RAs)			0.67	0.53			0.52		1.01
Acid-Reducing Agents (antacids)			N/A	N/A			N/A		
CYP2C8 substrates					1.23				
CYP2D6 Substrates		9.55		1.35					
CYP3A Substrates					1.45	0.68		0.92	0.98
P-gp Substrates					2.8	1	1.32	1.56	1.23
Strong Inducers	0.66		0.19	0.17	0.28	0.04	0.13	0.22	0.4
CYP1A2 inhibitors			1.39						
Strong CYP3A Inhibitors			6.79	5.54	3.57	8.43	4.81	1.24	1.09
P-gp inhibitors	1.45								
TKIs (FGFR)	Erdafitinib	Futibatinib	Infgratinib	Pemigatinib					
Strong Inducers	0.38	0.36	0.44	0.15					
CYP2C9 inhibitors	1.48								
Strong CYP3A Inhibitors	1.54	1.41	7.22	1.89					
Acid-Reducing Agents (PPIs)		1.05	0.55	0.92					
Acid-Reducing Agents (H2RAs)			N/A	1.03					
TKIs (KIT, PDGFRA)	Avapritinib	Ripretinib							

Figure 1: Heat map summarizing the drug–drug interaction (DDI) mechanisms and area under the concentration-time curve (AUC) changes for each oral anticancer drug (OAD) with

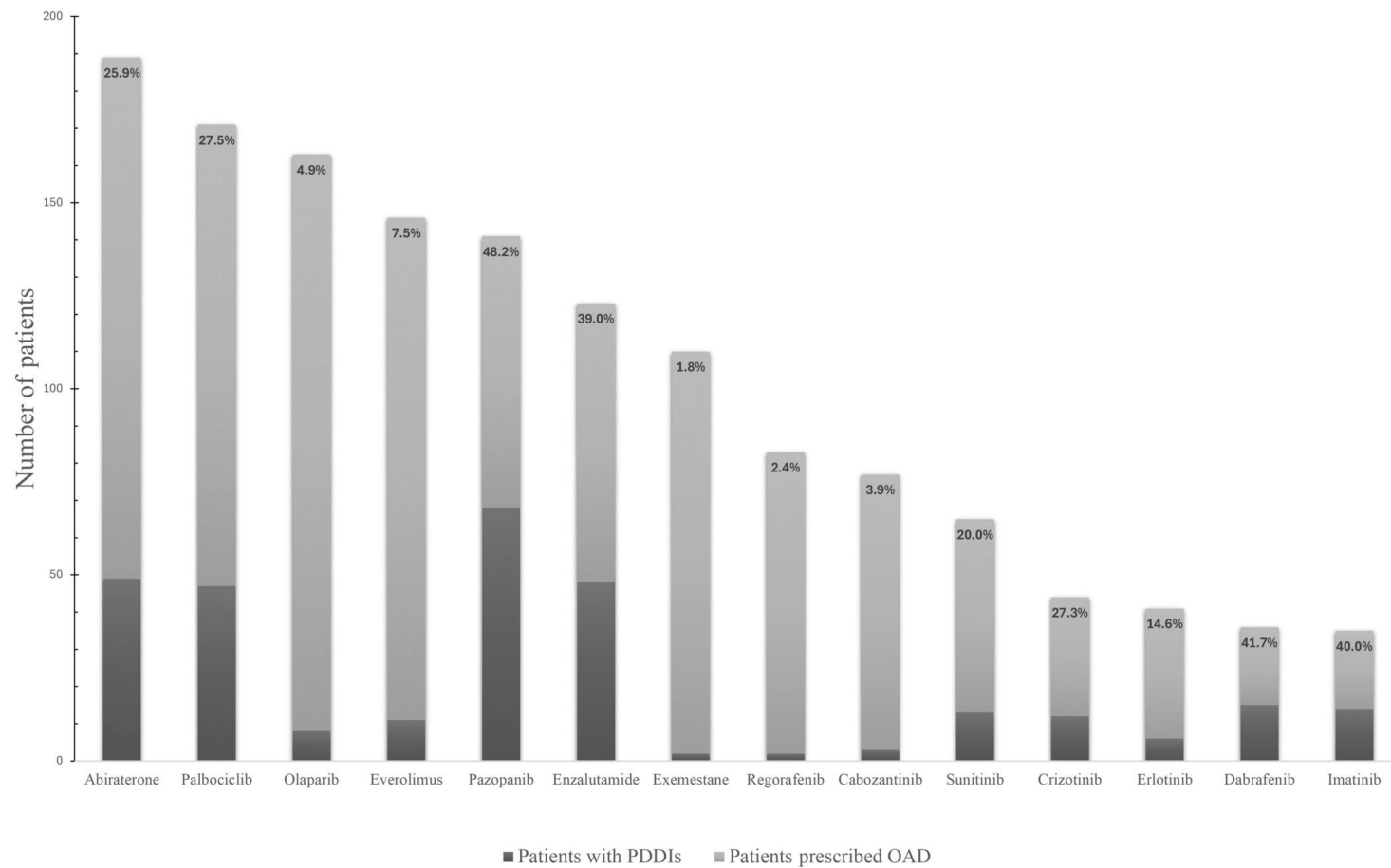


Figure 2: Patient-level potential drug–drug interaction (PDDI) prevalence for the most commonly prescribed oral anticancer drugs (OADs).